

Mohammadreza Maleki Ph.D.

Postdoctoral Fellow | Applied Machine Learning Researcher | Toronto, ON, Canada | +1 (647) 867-8247

[Academic Email](#) | [Personal Email](#) | [LinkedIn](#) | [GitHub](#) | [Website](#)

CV Highlights

- Postdoctoral researcher with a Ph.D. in Computer Systems Engineering; research in trustworthy AI, neural-network verification, robust training, and statistical change detection.
- Build reproducible Python/PyTorch research pipelines with benchmarking, ablations, error analysis, and robustness checks.
- Experience configuring and evaluating AI-agent workflows for analytics and content automation, using prompt refinement and structured validation to assess output quality and consistency.
- Author or coauthor of published and under-review papers in certified robustness, robust ensembles, and optimization-based verification at venues including IEEE SaTML, Canadian AI, AISTATS, and AAAI.

Employment History

Postdoctoral Fellow

Trustworthy AI Research Lab, Toronto Metropolitan University

Jun. 2026–Present

Toronto, ON, Canada

- Develop scalable certified-training and verification methods for neural networks using bound propagation, convex relaxations, and conic regularization.
- Design controlled experiments to evaluate accuracy, certified robustness, computational cost, stability, and sensitivity across model and perturbation settings.
- Collaborate on score-based quickest change detection with researchers at Princeton University and Augusta University, studying false-alarm behavior and detection-delay trade-offs.

Industry Research Collaborator

jamais51 Technologies Inc. and Toronto Metropolitan University

Jun. 2026–Present

Toronto, ON, Canada

- Formalize threat models, observability conditions, evidence provenance, and adversarial test cases for an independent verifier, with explicit certify/reject/abstain behavior.
- Translate research assumptions and failure modes into measurable acceptance criteria and reproducible validation experiments.

Head of Product & Data

ParagraphAI

Jun.–Sep. 2026

Toronto, ON, Canada

- Led product and data work across AI-enabled products, coordinating developers and domain experts around measurable requirements, experiments, and validated releases.
- Configured and optimized AI agents for analytics, social-media content generation, and business workflow automation; iteratively refined prompts and evaluation criteria to improve output quality and task performance.
- Designed controlled product experiments and analyzed PostHog, engagement, campaign, and revenue data to estimate effects and guide prioritization.
- On the SpeechLP project, collaborated with speech-language pathologists and developers on product planning, content verification, and experiments for speech-therapy games.

Doctoral Researcher / Research Assistant

Trustworthy AI Research Lab, Toronto Metropolitan University

Sep. 2021–May 2026

Toronto, ON, Canada

- Conducted research in neural-network verification and probabilistic robust ensembles, implementing and benchmarking certification methods on image-classification tasks.
- Led research from mathematical formulation and implementation through evaluation, publication, presentation, and mentorship.

Research Assistant

Analog Integrated Circuit Design Laboratory, University of Tehran

2017–2020

Tehran, Iran

- Conducted mathematical modeling, optimization, and simulation for wireless power and data-transfer systems and contributed to a national deep-brain-stimulator project.

Selected Projects

Scalable Neural Network Robustness Verification

- Developed sparse SOCP relaxations to strengthen certified robustness guarantees for ReLU networks; benchmarked fully connected and convolutional models on MNIST and CIFAR-10 against LP, SDP, and α, β -CROWN.
- Designed CORE, a class-wise verification cascade that matched SDP-certified robust accuracy while reducing average verification time by up to **86.30%** on a fully connected MNIST benchmark.

CoRe-CROWN: Conic-Regularized Certified Robust Training

- Developed certified robust training combining a Dual-CROWN objective with sparse SOCP-derived regularization applied during training.
- Designed influence-based block selection and local lifted constraints using McCormick envelopes and 2×2 PSD constraints; evaluated complete certified accuracy.

Sequential Distribution Shift Detection — DI-SCUSUM

- Co-developed training-free streaming detection with closed-form Gaussian-smoothed scores and KL-based detection-delay and false-alarm guarantees under empirical models.
- Evaluated Gaussian simulations, MNIST, and Oxford-IIIT Pet at matched false-alarm levels; achieved approximately **91% lower detection delay** than score-based CUSUM in a calibrated anisotropic Gaussian experiment, nearly matching likelihood-ratio CUSUM.

Secure Machine Learning Course Project — UN PET Lab Hackathon

- Applied missingness screening, histogram and correlation analysis, and SmartNoise queries to estimate hidden training labels under a privacy budget.
- Compared linear regression, random forests, and logistic regression; obtained a **0.6934 competition score** with logistic regression at total $\varepsilon = 1.85$. [Project code](#).

Education

Ph.D., Computer Systems Engineering

Toronto Metropolitan University; GPA: A^+

May 2026

Toronto, ON, Canada

Thesis: *Scalable Robust Verification of Neural Networks*; supervisor: Prof. Reza Samavi.

M.Sc., Electrical Engineering

University of Tehran; GPA: 17.29/20 (3.7/4.0)

Oct. 2020

Tehran, Iran

Thesis: *Phase Array Inductive Wireless Power Transfer for Moving Load*; supervisor: Prof. Shahin Jafarabadi Ashtiani.

B.Sc., Electrical Engineering – Communication

Isfahan University of Technology; GPA: 16.66/20 (3.67/4.0)

Aug. 2017

Isfahan, Iran

Capstone: *Basics and Applications of Digital Controlled Oscillators*; supervisor: Prof. Masood Omoomi.

Elective project: *Design and Simulate a Microwave Amplifier in the X Band*.

Publications and Manuscripts

- **M. Maleki**, D. Sadig, R. Sidibomma, M. Fazlyab, A. Adibi, and R. Samavi. *Certified Robustness via Sparse Second-Order Cone Relaxations*. Under review, AISTATS 2027.
- A. Adibi*, **M. Maleki***, S. Kulkarni, and H. V. Poor. *Quickest Change Detection with Diffusion-Integrated Scores*. Under review, AISTATS 2027. *Equal contribution.
- **M. Maleki** et al. *CoRe-CROWN: Conic-Regularized Certified Robust Training*. Under review, AAAI 2027.
- **M. Maleki** et al. *Cascading Robustness Verification: Toward Efficient Model-Agnostic Certification*. IEEE SaTML, 2026.
- D. Sadig, **M. Maleki**, H. Karimi, and R. Samavi. *CEAR: Certified Ensemble Adversarial Robustness in DNNs*. Canadian AI, 2026.
- S. Pahlavan et al., including **M. Maleki**. *Using Overlapped Resonators in Wireless Power Transfer for Uniform Electromagnetic Field and Removing Blank Spots in Free-Moving Applications*. Electronics, 2022.

Technical Skills

- **Programming and ML:** Python, SQL, MATLAB, C; PyTorch, TensorFlow, NumPy, pandas, SciPy, scikit-learn; familiarity with Hugging Face and JAX.
- **Agentic AI and evaluation:** AI-agent workflows, prompt design and optimization, structured evaluation, iterative testing, model/agent performance analysis, task-oriented automation.
- **Statistics and optimization:** probability, statistical inference, experimental design, causal inference, change detection, convex optimization, LP, SDP, SOCP.
- **Research workflow:** reproducibility, benchmarking, ablations, error analysis; Git/GitHub, Jupyter, Linux, LaTeX.